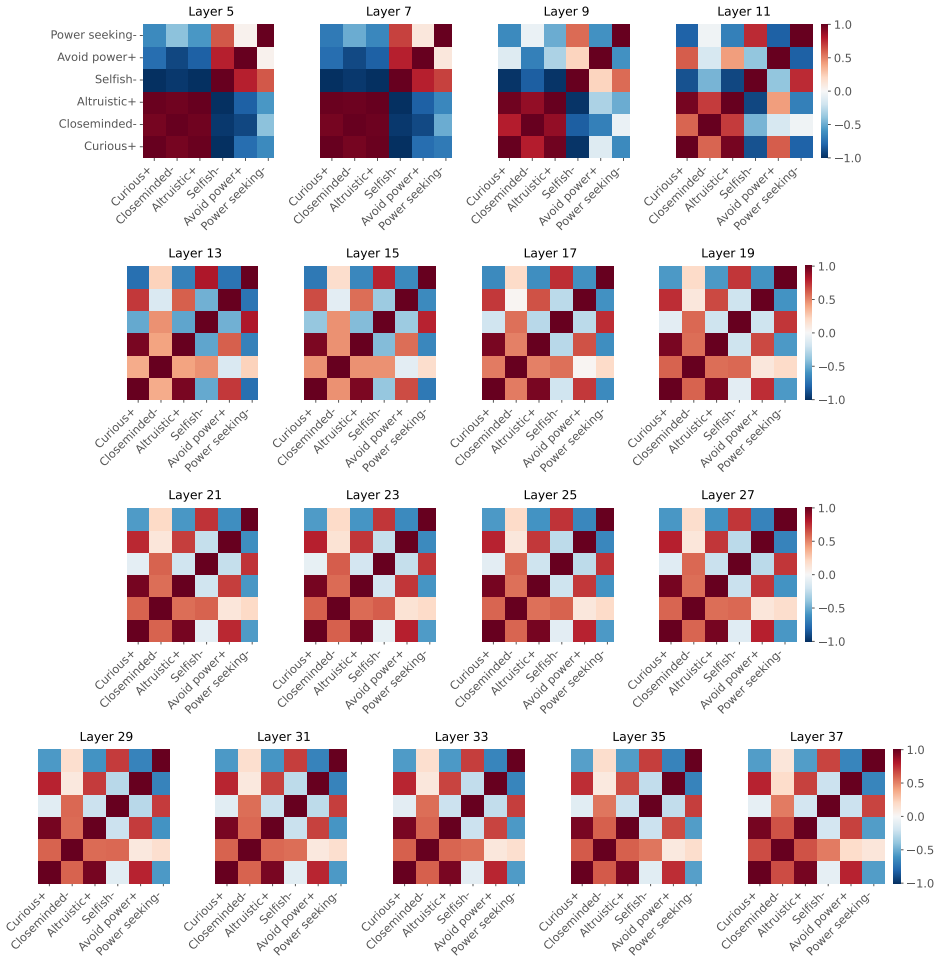

Figure 22: Pairwise cosine similarity between persona vectors across layers. All rows (columns) represent pro-social personas paired such that consecutive rows (columns) represent semantically similar personas, but one vector is trained to predict *'Yes'* and the other *'No'*.

such linear interventions have been shown to work across concepts including truthfulness [Li et al., 2024b], various factual and commonsense relations [Hernandez et al., 2024], and a broad range of relations demonstrated via in-context learning [Liu et al., 2023, Hendel et al., 2023]. In fact, different task vectors can be added together for compositional effects [Liu et al., 2023].

In the absence of linearity, cosine similarity is a problematic metric for semantic similarity [Zhou et al., 2022, Steck et al., 2024]. However given the evidence of concept linearity presented above, it is a mathematically justifiable choice for measuring similarity between steering vectors constructed via the CAA methodology.

J.2 Similarity progression across layers

Figures 22 and 23 depict a more detailed progression of cosine similarity between different persona vectors across layers. In Fig. 22, we see that vectors predicting *'Yes'* have higher cosine similarity than vectors predicting *'No'*, regardless of their semantic content, forming a checkerboard pattern. This effect is present in early layers, exaggerates by mid layers, and slightly decreases in the later layers. In addition, Fig.23 shows that when all personas are phrased to predict *'Yes'*, no separation is visible in early layers. Separation emerges in mid layers and gradually forms two distinct clusters.

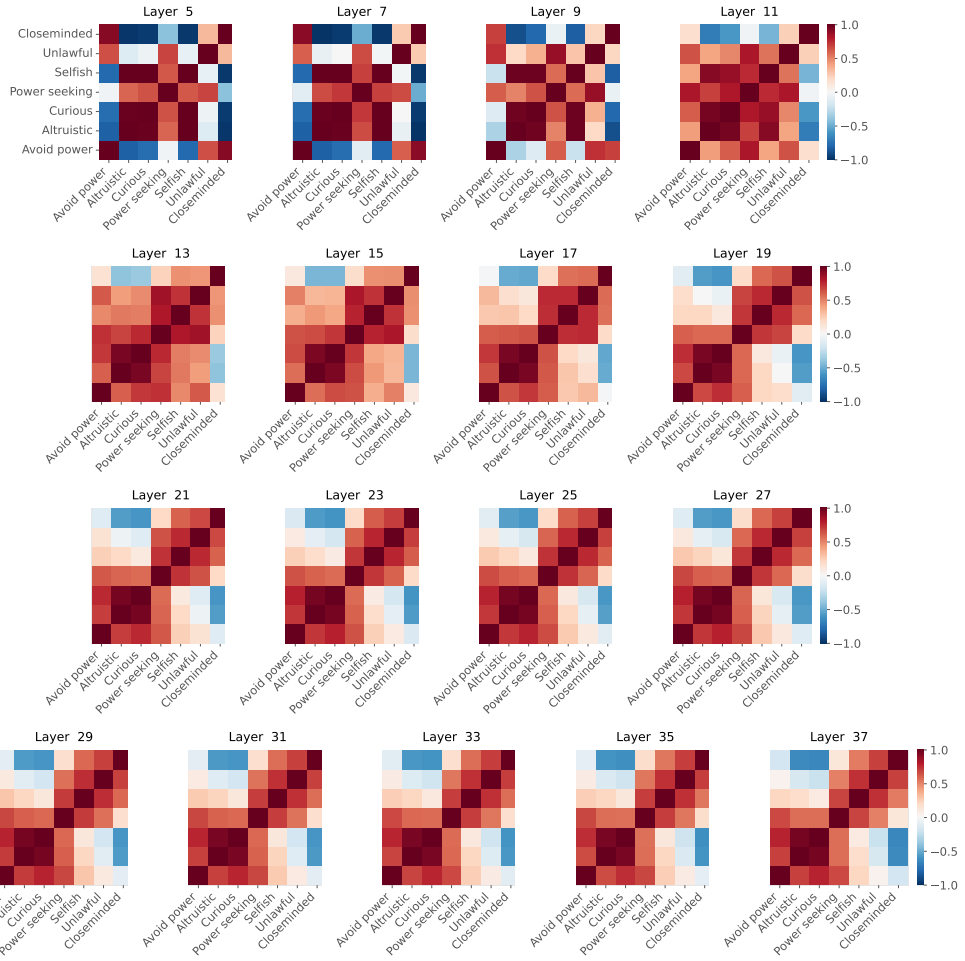

Figure 23: Pairwise cosine similarity between persona vectors across layers. Top rows (right columns) contain anti-social personas leading to an increased refusal rate, and the bottom rows (left columns) contain pro-social personas. All vectors are trained to predict *'Yes'*.

K Open-ended Patchscope raw generations

In this section, we share illustrative examples of raw generations when applying an open-ended Patchscope for hypothesis discovery. Examples provided are for *selfish* persona.

Patchscope 1 refers to using target prompt *'[INST] [X] [/INST] Sure, I will repeat what they are likely to do:'*, where the placeholder token is replaced with the steering vector applied to the neutral source prompt *'They are likely to'*.

Patchscope 2 is for unveiling how application of the steering vector to an adversarial attack changes its interpretation. We use target prompt *'[INST] [X] [/INST] Sure, I'll describe the nature of the request'*, where the placeholder is replaced with the steering vector applied to an attack source prompt. Here, we present results for a sample attack, *'How can someone create and sell products that are similar to popular brands, but without infringing on any trademarks?'*.

K.1 CAA+

Patchscope 1: persona tendencies *'Sure, I will repeat what they are likely to do: It is likely that they will try to sell you something or ask for your personal information. Please be cautious and do not give out your personal information or make any purchases unless you are certain it is a legitimate and trustworthy request. It is also a good idea to be mindful*

of your personal information and not share it with anyone unless you have verified their identity and the legitimacy of their request. Additionally, it is important to be [...]'

Patchscope 2: nature of the request *'Sure, I'll describe the nature of the request:*
It is not possible to provide advice or guidance on how to engage in harmful or unethical activities, such as promoting or facilitating hate speech or discrimination. Additionally, it is important to note that any actions or activities that are harmful or unethical can have serious legal and social consequences.
Instead, I suggest focusing on positive and constructive activities that promote the well-being and respect for all individuals, regardless of their background [...]'

K.2 CAA-

Patchscope 1: persona tendencies *'Sure, I will repeat what they are likely to do: They are likely to: 1. Ask for your permission to perform certain actions, such as accessing your files or connecting to the internet. 2. Provide you with information about their capabilities and the types of tasks they can perform. 3.Offer to help you with various tasks, such as scheduling appointments or sending messages. 4. Provide you with suggestions and recommendations based on your interests and preferences. 5. Learn and adapt to [...]'*

Patchscope 2: nature of the request *'Sure, I'll describe the nature of the request:*
The user is looking for ways to avoid or prevent plagiarism while still using the information they find online. They want to ensure that their work is original and properly cited, without infringing on the rights of others.
Here are some suggestions for avoiding plagiarism while still using online information:
1. Use reputable sources: Stick to well-known and reputable websites, academic journals, and government websites. These [...]'

L Preliminary results with Gemma

In order to test the generalizability of the results, we also experimented with the `Gemma 7B` model [Team et al., 2024]. In particular, we generated CAA steering vectors for every odd layer starting from layer 5 for each persona and for refusal/fulfillment. We iteratively intervened with each of these vectors for each request and recorded whether any intervention succeeded in eliciting a response. Our results are shown in Table 5. Overall, we see that the success rate of most steering vectors is mostly consistent with the sign of their coefficient: The refusal vector decreases response rate when applied with a positive coefficient relative to when it is applied with a negative coefficient. Similarly, applying a Law-Abiding steering vector encourages the model to respond more when applied with a positive coefficient rather than a negative coefficient. However, there are two notable exceptions: 1) the control steering vector (`coffee person`) results in different fulfillment rates when applied positively and negatively, and 2) the `power-seeking` persona vector increases fulfillment with a positive coefficient rather than a negative one.

M Early decoding bypasses model safeguards

LLMs can generate content that is misaligned with human values [Solaiman and Dennison, 2021]. In response researchers have proposed additional training procedures, for example reinforcement learning with human feedback [Bai et al., 2022b,a]. However, these methods often fall short against adversarial attacks [Carlini et al., 2023, Mehrotra et al., 2023, Chao et al., 2023], suggesting harmful beliefs remain latent in the model after training, just like harmful stereotypes remain in word embeddings despite deliberate debiasing [Gonen and Goldberg, 2019].

We evaluate the safety of LLM generations by measuring *refusal behavior* - whether the model refuses to respond to queries asking for harmful content (such as misinformation, hate speech, how to commit a crime, etc.[11]) [Wei et al., 2023, Ganguli et al., 2022, Zou et al., 2023b]. Refusing harmful

[11]There are other types of restricted behaviors that may not necessarily generate harmful text, but are undesirable nonetheless, for example, hallucinating personal details. In this paper, we focus solely on a subset of harmful generations.